

7. Project Documentation Phase

ComicCraft – AI Comic Story Creator Using Gemini Models

Recording the project concept, architecture, implementation, and usage

1. Documentation Purpose

The project documentation describes what ComicCraft does, how its different components work together, how to run the application locally, and how users follow the comic-generation workflow.

2. Project Summary

ComicCraft is a web-based AI comic creation application. Users enter a story prompt and preferences, after which the system produces a five-panel outline, narration and dialogue, illustrations, a preview, and a downloadable PDF.

3. Technology Stack

The project uses FastAPI and Uvicorn for the backend and server, HTML/CSS/Jinja2 for the interface, Gemini Flash and Gemini Pro for text generation, Hugging Face Diffusers with Stable Diffusion for image generation, and FPDF for PDF export.

4. Module Reference

`gemiini_flash.py`: `generate_outline()`. `gemiini_pro.py`: `generate_story()`.
`image_generator.py`: `generate_image()`. `layout_builder.py`: `build_comic_layout()`.
`exporters.py`: `save_pdf()`. `routes.py`: application routes and workflow integration.

5. Route Reference

`/` loads the homepage; `/generate` processes browser form submissions; `/generate-comic/json` handles JSON API requests; `/export-success` shows export confirmation; `/test-image` is used to test image generation.

6. User Guide

Open the local homepage, enter a story prompt and the required preferences, and submit the form. Review the generated comic on the preview page. Use the PDF download option to save the comic, and then use the success page to return and generate another comic.

7. Setup and Run Instructions

Create and activate a virtual environment, install the dependencies listed in `requirements.txt`, configure `GEMINI_API_KEY` and `HF_API_KEY`, and start the application with `uvicorn app.main:app --reload`. Open `127.0.0.1:8000` for the application and `/docs` for interactive API documentation.

8. Output and Storage

Generated panel illustrations are saved in `static/panels`, while exported PDF files are stored in `static/exports`. The guide specifies timestamp-based PDF filenames.

9. Scope and Future Enhancements

The current version is focused on generating a single comic and does not provide user account management. Possible future enhancements mentioned in the guide include user profiles, comic libraries, and multi-page story arcs.

10. Documentation Outcome

A complete project documentation set should contain the project overview, requirements, architecture, module descriptions, route information, setup instructions, user workflow, and testing observations. Any testing results included should be based on actual execution rather than assumptions.